Machine Learning Assisted Quantum State Tomography in a Photonic Polarisation System

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I. INTRODUCTION

A. Quantum state tomography and scaling limitations

Quantum state tomography (QST) is the task of reconstructing the quantum state of a system from measurement data obtained by performing repeated measurements on identically prepared copies of the system. In general, the state of a quantum system is described by a density matrix ρ, which is Hermitian, positive semidefinite, and has unit trace. These constraints define a highly structured subset of all complex matrices corresponding to physically allowed quantum states. For a system with n qubits, the density matrix contains $4^n - 1$ independent real parameters [1], implying that the amount of information required to fully characterise the state grows exponentially with system size. In this project, attention is restricted to a two-qubit system, for which quantum states are represented by a 4×4 density matrix.

In practical implementations of QST, measurements are described by a set of positive operator-valued measure (POVM) elements $\{E_k\}$, with outcome probabilities given by Born's rule,

$$p_k = \mathrm{Tr}(\rho E_k). \tag{1}$$

In many experimental settings, including photonic polarisation tomography, the POVM elements correspond to projectors onto specific measurement bases. Although individual measurement outcomes are inherently stochastic, repeated measurements allow the associated probabilities to be estimated. By performing measurements in a sufficiently large number of distinct bases, it is in principle possible to obtain an informationally complete dataset from which the underlying density matrix can be inferred.

However, the inversion of measurement data into a physical density matrix is a non-trivial task. Linear inversion methods, such as Stokes reconstruction, provide a fast and conceptually simple approach by directly solving the linear system defined by the measurement probabilities. While often reasonably accurate, such methods are highly sensitive to statistical fluctuations and experimental imperfections, which can easily lead to unphysical estimates. In particular, linear inversion does not enforce the positivity constraint on ρ, and even small errors in the estimated probabilities can result in density matrices with negative eigenvalues. Since physical quantum states must satisfy strict positivity and normalisation constraints, this limitation necessitates the use of more sophisticated reconstruction techniques [1].

A widely used approach to address this issue is maximum-likelihood estimation (MLE), in which one seeks the density matrix that maximises the likelihood of the observed measurement outcomes subject to the physical constraints on ρ [1]. MLE guarantees a physically valid reconstruction and provides a statistically well-motivated estimate of the state. However, this comes at the cost of solving a constrained, nonlinear optimisation problem whose computational expense grows rapidly with the dimension of the Hilbert space. As a result, MLE-based tomography can become prohibitively slow even for moderate system sizes, representing a bottleneck in experiments requiring high-throughput or near-real-time analysis.

These scaling and computational challenges motivate the exploration of alternative reconstruction approaches that maintain physical validity while reducing computational overhead. In particular, data-driven methods such as neural networks offer the possibility of learning an approximate inverse map from measurement data to quantum states, shifting the computational cost from reconstruction to an offline training phase. While such approaches do not eliminate the fundamental exponential scaling of tomography, they may offer practical advantages in speed and robustness for fixed system sizes. The application of machine learning techniques to quantum state tomography is discussed further in Section I C.

B. Photonic quantum tomography and measurement structure

Photonic systems play a central role in quantum information science, particularly in quantum communication, quantum metrology, and foundational tests of quantum mechanics. Photons are attractive carriers of quantum information due to their long coherence times, relative immunity to environmental decoherence, and precise controllability using standard optical components. In many optical experiments, qubits are encoded in the polarisation degree of freedom of photons, with the logical basis states identified as horizontal $|H\rangle$ and vertical $|V\rangle$ polarisations. This encoding provides a simple and well-understood physical realisation of a two-level quantum system.

Entangled two-qubit photonic states can be generated using spontaneous parametric down-conversion (SPDC) [2] or related nonlinear optical processes. In such experiments, photon pairs are produced whose joint polarisation state exhibits nonclassical correlations. Accurately characterising these states is essential for validating entanglement generation, benchmarking quantum devices, and verifying quantum communication protocols. As a result, photonic polarisation tomography has become a standard testbed for quantum state reconstruction methods.

In photonic tomography experiments, quantum state reconstruction is performed by measuring coincidence counts corresponding to a fixed set of projective measurements on each photon. Experimentally, these measurements are realised using combinations of half-wave plates, quarter-wave plates, and polarising beam splitters, which allow different polarisation measurement bases to be selected in a controlled and repeatable manner. Using such components, one can access a discrete set of single-qubit polarisation states,

$$\{|H\rangle, |V\rangle, |D\rangle, |A\rangle, |R\rangle, |L\rangle\},$$

where $|D\rangle$ and $|A\rangle$ correspond to diagonal and anti-

diagonal linear polarisations, and $|R\rangle$ and $|L\rangle$ correspond to right- and left-circular polarisations [3]. These states form three mutually unbiased bases, providing complementary information about the quantum state.

Mathematically, the single-qubit polarisation states can be written as

$$|H\rangle = \begin{bmatrix}1\\0\end{bmatrix}, \quad |V\rangle = \begin{bmatrix}0\\1\end{bmatrix},$$
$$|D\rangle = \tfrac{1}{\sqrt{2}}(|H\rangle + |V\rangle), \quad |A\rangle = \tfrac{1}{\sqrt{2}}(|H\rangle - |V\rangle),$$
$$|R\rangle = \tfrac{1}{\sqrt{2}}(|H\rangle + i|V\rangle), \quad |L\rangle = \tfrac{1}{\sqrt{2}}(|H\rangle - i|V\rangle),$$

with associated rank-one projectors $|i\rangle\langle i|$. Full two-qubit tomography is achieved by taking tensor products of these single-qubit projectors, resulting in a structured set of 36 two-qubit projectors. Together, these projectors form an informationally complete measurement collection, meaning that the corresponding measurement statistics contain sufficient information to reconstruct a general two-qubit density matrix.

Importantly, these measurement settings are not arbitrary mathematical choices. They are determined by the physical constraints of the optical setup and correspond to specific, reproducible orientations of wave plates and beam splitters, as implemented in established photonic tomography schemes such as those described by Wang *et al.* [2]. While any complete set of Hermitian operators could in principle be used for tomography, restricting attention to experimentally accessible projectors ensures that the reconstruction problem remains grounded in physically realisable measurements.

This structured measurement design has important implications for learning-based reconstruction methods. Measurement outcomes arise from a fixed set of projectors with known tensor-product structure, rather than from unstructured features. This creates opportunities to incorporate prior physical knowledge into reconstruction algorithms, while also placing limits on how assumptions such as locality in convolutional networks should be interpreted. Finally, photonic experiments are unavoidably subject to noise arising from misalignment of optical components, polarisation drift, detector dark counts, and background scattering [2]. These effects degrade reconstruction performance and motivate the noise models discussed in Section II.

C. Machine learning approaches to quantum state tomography

Recent years have seen growing interest in the application of machine learning (ML) techniques to quantum state tomography as a means of mitigating the computational and noise-related challenges associated with traditional reconstruction methods. In contrast to iterative optimisation-based approaches such as maximum-likelihood estimation (MLE), machine learning models can be trained to learn a direct mapping from measurement data to quantum states. Once trained, such models can perform reconstructions extremely rapidly, shifting much of the computational cost from reconstruction to an offline training phase.

From a conceptual perspective, applying ML to quantum state tomography is nontrivial. The mapping from measurement probabilities to density matrices is highly structured, constrained, and nonlinear, and the space of physically valid quantum states occupies only a small subset of all complex matrices. As a result, naïvely applying generic machine learning models does not automatically guarantee physically meaningful reconstructions. Nevertheless, ML methods offer a flexible framework for approximating complex inverse mappings, particularly in settings involving noise, incomplete data, or experimental imperfections.

A variety of ML approaches to QST have been proposed, including feedforward neural networks, convolutional neural networks (CNNs), and recurrent neural networks (RNNs). These methods have shown promising performance in numerical simulations and proof-of-principle studies, particularly when measurement data is noisy or limited. In many cases, learning-based approaches have been found to offer improved robustness to certain noise sources while providing substantial gains in reconstruction speed once trained.

Of particular relevance to this project is the work of Lohani *et al.*, who introduced a CNN-based approach to quantum state reconstruction from noisy tomographic measurements. Their method arranges measurement probabilities into a structured grid reflecting the tensor-product measurement scheme and trains the network to output a parametrisation of the density matrix that enforces physical constraints by construction. Their results demonstrate that neural networks can achieve reconstruction fidelities comparable to traditional methods while offering orders-of-magnitude reductions in inference time.

Despite these successes, existing learning-based tomography approaches often rely on generic neural network architectures and design choices inherited from classical data domains such as image processing. In particular, CNNs are typically employed using standard kernels that assume locality and translation invariance, assumptions that may not naturally align with tomographic measurement structure. Similarly, training objectives such as mean-squared error are frequently adopted for convenience, despite not directly reflecting physically meaningful notions of state similarity.

These observations motivate a more careful examination of how physical structure and prior knowledge can be incorporated into learning-based reconstruction schemes. Rather than treating tomography as a generic regression problem, one may ask how the known properties of quantum measurements, state-space geometry, and experimental constraints can inform the choice of network architecture, loss function, and data representation. Addressing these questions is central to the present project.

D. Physical constraints and Cholesky-based parametrisation

A key challenge in applying neural networks to quantum state tomography is ensuring that reconstructed states satisfy the physical constraints required of a density matrix. A valid quantum state must be Hermitian, positive semidefinite, and have unit trace. As with Stokes reconstruction, directly training a network to output the elements of ρ does not guarantee that these constraints are satisfied. Small prediction errors can therefore lead to unphysical states with negative eigenvalues or incorrect normalisation. While post-processing steps can partially enforce physicality, they may introduce distortions that are difficult to control.

One principled approach to this problem is to parametrise

the density matrix in a way that enforces physical constraints by construction. In this work, a Cholesky-based parametrisation is employed, in which the density matrix is written as

$$\rho = \frac{\tau^\dagger \tau}{\mathrm{Tr}(\tau^\dagger \tau)}, \tag{2}$$

where τ is a lower-triangular complex matrix. Since $\tau^\dagger \tau$ is positive semidefinite for any choice of τ, and the explicit normalisation enforces unit trace, this parametrisation guarantees that ρ is a valid physical density matrix independent of the values predicted for the elements of τ [3].

For a two-qubit system, the density matrix occupies a 15-dimensional real parameter space once the trace constraint is enforced. The matrix τ provides a convenient way to represent this space using a fixed set of real parameters corresponding to the real and imaginary parts of its non-zero entries. While this representation is slightly overparameterised prior to normalisation, it defines a smooth and unconstrained mapping from a real-valued parameter vector to the space of physical quantum states, making it well suited for neural network outputs.

From a learning perspective, the Cholesky parametrisation alters the reconstruction task by enforcing physical constraints analytically rather than implicitly. The network is free to focus on learning the inverse mapping from measurement data to state parameters without being penalised for intermediate unphysical predictions. This contrasts with heuristic approaches such as eigenvalue clipping, which can distort reconstructed states, particularly near the boundary of state space. For these reasons, the Cholesky parametrisation is adopted throughout this work and forms a central component of the reconstruction pipeline.

E. Motivation and scope of this project

While previous work has demonstrated the potential of neural networks for quantum state tomography, relatively little attention has been paid to the role of physical structure in network design. In particular, convolutional neural networks are often applied using generic kernel designs inherited from image processing tasks, despite the fact that tomographic measurement data arises from a fixed set of physical projectors rather than spatially local image features. This raises the question of whether such architectures are well matched to the underlying reconstruction problem.

In photonic quantum tomography experiments, measurement outcomes are directly associated with specific tensor-product projectors determined by the experimental apparatus. This structure is known *a priori* and constrained by the physics of the system and available optical components. Incorporating this knowledge into learning-based reconstruction methods may improve performance, data efficiency, or robustness to experimental imperfections. Conversely, ignoring it may limit the ability of generic architectures to extract physically meaningful information from tomographic data.

The primary aim of this project is therefore not to demonstrate immediate superiority of learning-based reconstruction over classical methods, but to establish a clear and reproducible baseline for neural network–based quantum state tomography within a physically motivated framework. This includes defining a consistent physical model, measurement scheme, noise model, and reconstruction pipeline, and benchmarking standard fully connected and convolutional architectures against classical techniques.

As an interim report, this work focuses on documenting the reconstruction pipeline and assessing baseline performance under controlled conditions, rather than on exhaustive optimisation or large-scale architectural exploration. The results are intended to clarify both the capabilities and limitations of generic learning-based approaches and to identify key challenges that must be addressed in future work, including the development of explicitly physics-informed neural network architectures.

II. METHODOLOGY

This section describes the physical model, data generation procedure, and reconstruction methods used to benchmark classical and neural network–based quantum state tomography techniques. Although the simulations are inspired by photonic polarisation experiments, all results presented here are obtained using an abstract two-qubit model defined directly at the level of density matrices and projective measurements. This allows the reconstruction problem to be studied in a controlled and reproducible setting while retaining a clear connection to experimentally relevant measurement schemes.

A key aim of this interim report is to document the full "lifecycle" of the reconstruction problem. This includes generating physically valid quantum states, constructing an informationally complete set of measurement projectors, simulating measurement noise motivated by experimental imperfections, and applying both classical and learning-based reconstruction methods. By separating these stages, the impact of individual modelling choices can be isolated and different reconstruction strategies can be compared on equal footing.

Emphasis is placed on establishing a robust baseline rather than exhaustive optimisation. The methodology is therefore described in sufficient detail for reproducibility and to clarify which components of the pipeline are fixed and which are intended as tunable elements for future work. The present framework is intended to support later extensions, such as incorporating more realistic noise models, modifying neural architectures to better reflect measurement physics, or exploring alternative training objectives.

At this stage, several aspects of a full experimental tomography pipeline are deliberately simplified or excluded. In particular, the measurement data is represented at the probability level rather than as finite coincidence counts, and only a single class of measurement imperfection (operator mis-rotation) is modelled explicitly. These choices are made to isolate the effect of miscalibrated measurements and to establish a clear baseline before introducing additional sources of variability such as shot noise, detector dark counts, or state preparation errors. Throughout this section, modelling choices are therefore motivated both by their physical relevance and by their role in producing a modular benchmark that can be extended systematically.

A. Quantum states and parametrisation

The system under study consists of two qubits, whose quantum state is described by a 4×4 density matrix ρ acting on the Hilbert space $\mathcal{H} = \mathbb{C}^2 \otimes \mathbb{C}^2$. A two-qubit density matrix must be Hermitian, positive semidefinite, and have unit trace. These conditions restrict the state space to a 15-dimensional real manifold, corresponding to the $4^2 - 1$ independent real parameters required to specify a general two-qubit state.

To generate physically valid target states for training and evaluation, random mixed states are constructed using a Ginibre ensemble [3]. Specifically, a complex random matrix G with independent, normally distributed entries is generated and the corresponding density matrix is formed as

$$\rho = \frac{GG^\dagger}{\mathrm{Tr}(GG^\dagger)}. \tag{3}$$

This guarantees that ρ is Hermitian, positive semidefinite, and trace normalised by construction.

The Ginibre construction produces a broad distribution of mixed states with varying eigenvalue spectra, spanning a wide range of purities from nearly pure states (near the boundary of state space) to highly mixed states in the interior. This diversity is useful for benchmarking reconstruction algorithms, since performance can depend on purity. Nearly pure states are often more challenging due to their proximity to the positivity boundary, while highly mixed states tend to be more robust.

Training and testing on mixed states is also physically motivated. In realistic photonic experiments, pure states are rarely realised exactly: imperfect mode overlap, background counts, partial photon distinguishability, and decoherence effects all contribute to mixedness. Although the Ginibre ensemble does not model a specific source mechanism, it provides a practical and unbiased way to explore the two-qubit state space at this stage.

While the Ginibre ensemble is suitable as a general benchmark, it does not reflect the statistics of a specific experimental source. Future work may therefore restrict the state ensemble to physically motivated families (e.g. Bell-diagonal states or states typical of a given SPDC source). For the purposes of this interim report, the Ginibre ensemble provides a convenient baseline distribution.

It is also worth noting that using a broad, "source-agnostic" ensemble is a deliberate choice for establishing a baseline. If the training distribution were restricted too early to a narrow family of states, a learning-based model could appear to perform well while effectively specialising to that family. By contrast, the Ginibre ensemble forces the reconstruction methods to cope with a wide range of eigenvalue spectra and coherence patterns. This makes improvements (for example from more physically informed kernels or loss functions) easier to interpret, because the benchmark is not overly tailored to a particular state class.

To ensure that reconstructed states produced by neural networks are also physical, a Cholesky-based parametrisation is employed. The density matrix is expressed as

$$\rho = \frac{\tau^\dagger \tau}{\mathrm{Tr}(\tau^\dagger \tau)}, \tag{4}$$

where τ is a lower-triangular complex matrix. By construction, $\tau^\dagger \tau$ is positive semidefinite and the trace normalisation enforces $\mathrm{Tr}(\rho) = 1$, so any choice of parameters defining τ yields a valid density matrix.

For a two-qubit system, τ contains a fixed number of independent real parameters sufficient to represent the full state space once normalisation is imposed. In practice, the network predicts a real-valued parameter vector encoding the real and imaginary components of the non-zero elements of τ. This provides a smooth, unconstrained mapping from network output to physical density matrices.

Using τ as the network output changes the learning problem from predicting a constrained object to predicting unconstrained real parameters that are deterministically mapped into a valid state. This improves training stability by avoiding unphysical regions, removes the need for heuristic fixes such as eigenvalue clipping, and ensures all reconstructed states are physical by construction.

From a practical viewpoint, this parametrisation also provides a clean interface between classical and learning-based methods in the pipeline. Classical baselines can be evaluated on the same fidelity metric without requiring additional physicality enforcement steps to be comparable with the neural reconstructions. Moreover, because the map $\tau \mapsto \rho$ is differentiable, it can be included inside the training loop, allowing gradients to backpropagate through the physicality enforcement step rather than treating it as a post-processing operation. For these reasons, the Cholesky parametrisation forms a central component of the methodology used throughout this work.

B. Tomographic measurement model

Quantum state tomography is performed using a fixed set of projective measurements motivated by photonic polarisation experiments. For a single qubit, the experimentally accessible measurement bases correspond to the polarisation states

$$\{|H\rangle, |V\rangle, |D\rangle, |A\rangle, |R\rangle, |L\rangle\},$$

with associated rank-one projectors $|i\rangle\langle i|$. These states form three mutually unbiased bases (horizontal/vertical, diagonal/anti-diagonal, and right/left circular), which together provide sufficient information to reconstruct an arbitrary single-qubit state.

Full two-qubit tomography is achieved by forming tensor products of these single-qubit projectors, resulting in a set of $6 \times 6 = 36$ two-qubit projectors

$$P_{ij} = |i\rangle\langle i| \otimes |j\rangle\langle j|. \tag{5}$$

This set is informationally complete for two-qubit tomography: the corresponding measurement statistics contain sufficient information to reconstruct all $4^2 - 1 = 15$ independent real parameters of a general two-qubit density matrix. The measurement set is overcomplete, which is common in experiments and can improve robustness to noise.

The 36 projectors can be arranged as a 6×6 grid, where the row index corresponds to the projector applied on the first qubit and the column index corresponds to the projector applied on the second qubit. This structure is used to define the CNN input: the CNN receives a single-channel "image" of size 6×6, while the fully connected (FC) model receives the same values flattened into a length-36 feature

vector. This enables a controlled comparison between treating measurement outcomes as an unordered feature vector versus structured grid data.

The 6×6 grid ordering is a modelling choice rather than a physical requirement. Unlike image data, locality in this grid depends on how the single-qubit bases are arranged. Consequently, generic convolutional kernels may not align naturally with physically meaningful relationships between measurement operators. This motivates future work on more physically informed architectures, where kernel structure or connectivity is designed to reflect known relationships between projectors.

In the context of this interim report, the CNN should therefore be viewed primarily as a baseline diagnostic: it tests whether standard convolutional inductive biases (local receptive fields and weight sharing) provide any benefit when the input is arranged in a convenient grid. If the CNN underperforms relative to a fully connected model, this does not necessarily imply that convolutional structure is inappropriate in principle, but rather that the assumed notion of locality is not well matched to the measurement geometry. This framing is important because it clarifies that future improvements are expected to come from aligning the network architecture with the measurement physics, rather than from increasing model capacity alone.

It is also important to distinguish the full set of 36 projectors from a single projective measurement setting. In a physical experiment, one selects a pair of single-qubit bases (one per qubit), which defines a measurement setting corresponding to four orthogonal two-qubit outcomes that sum to the identity. Each such setting constitutes a projective POVM. Tomography proceeds by repeating measurements across multiple settings so that the union of all outcomes is informationally complete.

In this work, the 36 projectors are treated collectively as an informationally complete measurement set. This abstraction allows the reconstruction problem to be formulated cleanly at the level of probabilities, without explicitly modelling the scheduling of measurement settings. While simplified, it retains the essential structure of photonic polarisation tomography and supports direct method comparisons.

Given a density matrix ρ, the ideal probability associated with projector P_{ij} is

$$p_{ij} = \mathrm{Tr}(\rho P_{ij}). \tag{6}$$

In a real experiment, these probabilities would be estimated from coincidence counts accumulated over many repetitions. Here the data is represented directly at the probability level to isolate the effect of operator-level noise (described below) without introducing additional variance from finite counting statistics.

This probability-level representation has an additional benefit: it ensures that differences in performance between reconstruction methods are not obscured by stochasticity in the dataset generation. In particular, for a fixed ρ and a fixed set of noisy projectors, the measurement probabilities are deterministic. This makes the benchmarking results easier to reproduce and interpret at the interim stage. Incorporating explicit count sampling (and therefore shot noise) is a natural extension of the framework, but is deferred so that the current baseline focuses on calibration-like measurement imperfections.

C. Noise model

To model experimental imperfections commonly encountered in photonic polarisation tomography, noise is introduced at the level of the measurement operators rather than the quantum state. Each ideal projector P_{ij} is independently perturbed by applying a local unitary rotation,

$$P_{ij}^{\mathrm{noisy}} = U P_{ij} U^{\dagger}, \tag{7}$$

where $U = U_1 \otimes U_2$ and $U_1, U_2 \in \mathrm{SU}(2)$ are single-qubit rotations. The rotation parameters are sampled from Gaussian distributions with zero mean and standard deviation σ, which controls the overall noise strength and sets the typical deviation between intended and realised measurement operators.

Physically, this corresponds to implementing a slightly rotated measurement basis compared to the ideal one. In photonic experiments, such mis-rotations can arise from wave plate angle errors, polarisation drift in fibres, imperfect alignment, or residual calibration error. These effects are naturally captured by local $\mathrm{SU}(2)$ rotations acting on the measurement basis.

Applying noise at the projector level reflects the assumption that dominant error arises from measurement imperfections rather than decoherence of the quantum state prior to measurement. This distinguishes the model from state-noise models, where the system is assumed to undergo a noisy quantum channel before measurement. In photonic tomography, miscalibrated measurements can be a major source of reconstruction error, so modelling operator noise provides a useful test case.

In the present implementation, each projector is perturbed independently, producing a conservative noise model in which different measurement settings experience uncorrelated misalignments. While real experiments may exhibit correlated drifts, the independent-rotation model provides a simple and flexible way to generate a broad class of measurement imperfections without introducing additional assumptions.

Once the noisy projectors have been constructed, measurement probabilities are computed according to

$$p_{ij}^{\mathrm{noisy}} = \mathrm{Tr}(\rho P_{ij}^{\mathrm{noisy}}), \tag{8}$$

preserving Born's rule while incorporating the effect of misaligned measurement operators. In this interim-stage study, probabilities are used directly so that the impact of operator-level noise can be examined without the added variance introduced by finite sampling.

The parameter σ provides a single-parameter control over noise strength. For small σ, the noisy projectors remain close to ideal and reconstruction is expected to be accurate; as σ increases, the measurement operators deviate more substantially and reconstruction becomes more challenging. In this interim report, σ is fixed (e.g. $\sigma = 0.564$ for the results shown) to provide a consistent benchmark for comparing methods.

Fixing σ in this way is also a fairness choice: it ensures that all reconstruction methods are tested under the same measurement-imperfection regime, so observed differences in fidelity are attributable to the reconstruction strategy rather than to changes in dataset difficulty. In future work, scanning across σ would provide a more complete performance profile and could reveal regimes in which learning-

based methods become comparatively more robust. For example, one might expect the advantage of physics-informed inductive biases to become more apparent as calibration error increases and the inversion becomes less stable.

This noise model does not capture all experimental error sources, such as finite counting statistics, detector dark counts, or state preparation errors. These effects can be incorporated in future extensions. Nevertheless, the present model captures a physically meaningful and practically relevant class of imperfections and provides a useful testbed for benchmarking robustness to miscalibrated measurements.

D. Dataset construction

For each randomly generated density matrix, noisy measurement probabilities are evaluated for all 36 projectors and arranged into a 6×6 probability matrix M, with entries $M_{ij} = p_{ij}^{\text{noisy}}$. For fully connected models, the matrix is flattened into a 36-dimensional real-valued input vector $X \in \mathbb{R}^{36}$. For convolutional models, the 6×6 structure is retained and treated as a single-channel input, allowing the network to exploit any local structure induced by the chosen grid ordering.

The target output Y does not consist of the elements of ρ directly. Instead, the target is a real-valued parameter vector encoding the complex lower-triangular matrix τ, with ρ reconstructed via $\rho = \tau^\dagger\tau/\mathrm{Tr}(\tau^\dagger\tau)$. In practice, the network outputs a 16-dimensional real-valued vector which is mapped deterministically onto the real and imaginary components of the non-zero elements of τ. This slightly overparameterised representation simplifies implementation while maintaining a smooth mapping to physical density matrices.

This input/output choice reflects the design philosophy of the pipeline: inputs correspond directly to measurement outcomes (here at the probability level), while outputs enforce physical constraints analytically through the deterministic map $\tau \mapsto \rho$. This allows a clear distinction between reconstruction methods that enforce physicality implicitly and those that enforce it by construction. The representation also aligns closely with Lohani *et al.* [3], making the present implementation a baseline emulation of their setting.

Datasets are constructed by sampling N independent quantum states and computing the corresponding noisy measurement matrices. Results are reported for a range of dataset sizes to examine how performance scales with available training data. Each dataset is divided into training and test subsets, and all reported metrics are evaluated on held-out test data to assess generalisation.

Varying N probes the data efficiency of the models: methods that achieve high fidelity with small N are more practical in settings where labelled data is costly, while methods requiring large training sets may be less suitable for near-term deployment.

Finally, the measurement and noise models define a specific distribution over inputs M. Networks trained on one such distribution (fixed σ and state ensemble) are not guaranteed to generalise to substantially different experimental conditions without retraining. This type of distribution shift is particularly relevant in experimental contexts, where drift, recalibration, or changes in the state source can alter the statistics of the measurement data. A natural extension of the present baseline is therefore to test robustness to changes in σ or to train on mixtures of noise strengths, which could encourage the network to learn a more broadly applicable inverse map. In the interim report, however, the focus is deliberately on controlled benchmarking within a fixed setting to establish a clear reference point.

E. Classical baseline: linear inversion (Stokes reconstruction)

As a classical benchmark, a linear inversion method based on Stokes reconstruction is implemented. Measurement probabilities are related linearly to the elements of ρ through the known set of projectors. By vectorising ρ and expressing each probability as a linear functional of its elements, the reconstruction can be written as an overdetermined linear system,

$$\mathbf{p} = \mathcal{A}\,\mathrm{vec}(\rho), \tag{9}$$

where $\mathbf{p}$ is the vector of probabilities and $\mathcal{A}$ is a measurement matrix constructed from the projectors $\{P_{ij}\}$. A least-squares solution provides an estimate of the density matrix.

In practice, linear inversion does not automatically satisfy all physical constraints. Hermiticity can be enforced by symmetrisation, but positivity is not guaranteed and noise can lead to negative eigenvalues. In the present implementation, Hermiticity is enforced and the trace is renormalised, but no additional constraint is imposed to ensure positivity. The reconstructed matrix may therefore be unphysical, particularly in the presence of noise.

Despite this limitation, Stokes reconstruction is computationally efficient and widely used for low-dimensional systems. It requires only a single matrix inversion or least-squares solve and is far faster than iterative optimisation approaches such as MLE. For two-qubit systems the inversion problem is small and often yields high-fidelity estimates even under moderate noise.

In this interim report, Stokes reconstruction serves as a strong, transparent baseline. Because it directly inverts the known measurement model, it does not depend on training data or generalisation and is therefore particularly effective in the regime studied here. In this sense, it is also a deliberately difficult baseline to beat in the low-dimensional setting, and learning-based methods are not expected to outperform it without incorporating additional physical structure or being tested in more challenging regimes. Its limitations motivate the broader aims of the project: as system size grows or measurement imperfections increase, lack of enforced positivity and sensitivity to noise become more problematic, while constrained optimisation methods become expensive. Learning-based approaches offer a potential route to combine physical validity with fast inference, but likely require inductive biases informed by measurement physics. Including Stokes reconstruction therefore provides a clear reference point for evaluating future physics-informed architectures.

F. Neural network reconstruction methods

Two neural network architectures are considered: a fully connected (FC) feedforward network and a convolutional neural network (CNN). In both cases, the input consists of tomographic measurement data derived from the 36 two-qubit projectors. For the FC model, the 6×6 probability matrix is flattened into a 36-dimensional vector. For the CNN, the same data is reshaped into a 6×6 single-channel array, preserving the grid structure induced by the tensor-product measurement scheme.

In both architectures, the network outputs real-valued parameters encoding the Cholesky matrix τ, and the reconstructed density matrix is obtained analytically via $\rho = \tau^\dagger\tau/\mathrm{Tr}(\tau^\dagger\tau)$. This ensures that all reconstructed states are physical by construction and shifts the learning task toward approximating the inverse mapping from measurement statistics to state parameters.

Although the FC and CNN models receive the same information, they embody different inductive biases. The FC network treats the input as an unordered feature vector and can model arbitrary global correlations, but does not exploit assumed structure. The CNN assumes that nearby entries in the 6×6 grid may contain related information and that shared convolutional filters can capture these relationships.

In conventional image tasks, this is justified by spatial locality and translation invariance. In tomography, locality is more subtle: the 6×6 grid reflects an ordering of tensor-product projectors rather than a physical spatial arrangement. Whether neighbouring entries correspond to physically similar measurements depends on basis ordering and measurement structure. As a result, generic convolutional kernels may not align with physically meaningful relationships between projectors.

In the interim-report context, this motivates treating the CNN as a baseline that tests a common modelling approach rather than as an optimised architecture. If performance is limited, the conclusion is not that convolution is inherently unsuitable, but that the naive notion of locality is unlikely to be the correct inductive bias. This directly motivates the project's future direction: defining kernels or receptive fields that reflect the measurement process, for example by coupling outcomes that share the same basis on one subsystem, or by encoding known symmetries between projectors.

Networks are trained using the Adam optimiser with mini-batch gradient descent. Two loss functions are investigated: a mean-squared error (MSE) loss applied to reconstructed density matrix elements, and a fidelity-based loss defined as $1 - \mathcal{F}(\rho_{\mathrm{true}}, \rho_{\mathrm{pred}})$, where $\mathcal{F}$ is the Uhlmann fidelity. Fidelity is basis independent and physically meaningful, making it a natural objective for state reconstruction. However, because it is nonlinear and involves matrix square roots, it can lead to a more complex loss landscape than MSE. Comparing these objectives provides insight into the trade-offs between optimisation simplicity and physical relevance.

Training is performed offline: parameters are optimised on a training dataset and evaluated on held-out test data. Once trained, the networks perform reconstruction via a single forward pass, in contrast to optimisation-based methods such as MLE which require an iterative procedure for each new reconstruction. Although inference time is not the primary metric in this interim report, it remains a key practical motivation for learning-based approaches.

G. Loss functions and the role of fidelity-based training

Both mean-squared error (MSE)–based training and fidelity-based training are considered to investigate how the choice of objective affects reconstruction quality, stability, and physical relevance. MSE penalises element-wise deviations between predicted and target quantities and encourages the network to minimise average squared differences between corresponding matrix entries.

However, MSE does not necessarily align with physically meaningful notions of state similarity. In quantum information, the standard figure of merit is typically the Uhlmann fidelity, which quantifies overlap and distinguishability between density matrices. Fidelity is invariant under unitary transformations and depends on the full operator structure of the states rather than a particular matrix representation, making it a more appropriate measure of reconstruction quality.

A limitation of MSE-based training is that small element-wise errors do not always imply small physical errors. Correlated deviations can produce significant changes in eigenvalues or coherences, especially for states near the boundary of state space. Two density matrices may therefore exhibit a small MSE while corresponding to noticeably different physical states.

To address this, a fidelity-based loss is also considered,

$$\mathcal{L}_{\mathrm{fid}} = 1 - \mathcal{F}(\rho_{\mathrm{true}}, \rho_{\mathrm{pred}}), \tag{10}$$

where $\mathcal{F}$ is the Uhlmann fidelity. Training against infidelity aligns the optimisation objective with the evaluation metric reported in the results and encourages the network to prioritise physically relevant corrections.

Fidelity-based losses introduce additional optimisation complexity: fidelity involves matrix square roots and nonlinear eigenvalue-dependent structure, which can lead to a more intricate loss landscape and noisier gradients than MSE. As a result, training with a fidelity-based objective may require more careful tuning of learning rates, regularisation, and optimisation schedules, particularly for convolutional models. Nevertheless, fidelity-based training is an important benchmark objective in quantum state reconstruction. In the interim report, it is treated as a principled baseline against which future improvements (for example hybrid objectives or physics-informed regularisation) can be evaluated.

H. Evaluation metrics

Reconstruction performance is quantified using the Uhlmann fidelity between the true density matrix ρ_{true} and the reconstructed density matrix ρ_{pred}. Fidelity provides a standard, physically meaningful measure of similarity between quantum states: $\mathcal{F} = 1$ corresponds to perfect reconstruction, while lower values indicate increasing deviation. Fidelity is widely used because it is invariant under unitary transformations and closely reflects state distinguishability.

For each reconstruction method and training configuration, performance is reported as the average test-set fidelity. Fidelities are computed for each state in a held-out test set and averaged to produce a single summary statistic. Results are presented as a function of the training dataset size

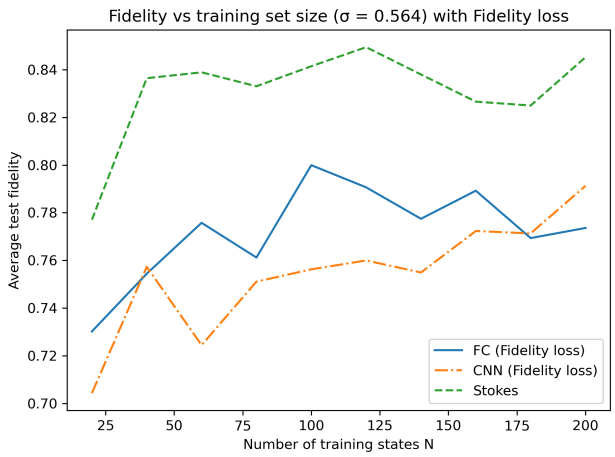

FIG. 1: Average test-set Uhlmann fidelity as a function of training set size N at fixed measurement-noise strength $\sigma = 0.564$, for the fully connected (FC) and convolutional neural network (CNN) reconstructions trained using a fidelity-based loss, $\mathcal{L} = 1 - \mathcal{F}(\rho_{\text{true}}, \rho_{\text{pred}})$. The Stokes linear-inversion baseline is shown for comparison.

FIG. 2: Average test-set Uhlmann fidelity as a function of training set size N at fixed measurement-noise strength $\sigma = 0.564$, for FC and CNN reconstructions trained using a mean-squared error (MSE) loss on the reconstructed density matrix elements. The Stokes linear-inversion baseline is shown for comparison.

N, enabling direct comparison between classical and neural network–based methods across different data regimes. Reporting performance as a function of N provides insight into data efficiency.

While mean fidelity is the primary metric, it does not capture all aspects of reconstruction behaviour. For this reason, qualitative features such as stability across dataset sizes, sensitivity to loss function choice, and consistency between independent training runs are also considered when interpreting results. This emphasis is appropriate for an interim report: the goal is not to claim optimal performance, but to establish a reproducible benchmarking framework and identify limitations that motivate targeted future improvements in architecture, training strategy, and physical modelling assumptions.

III. CURRENT RESULTS AND FUTURE WORK

A. Reconstruction performance

The performance of the reconstruction methods described in Section II was evaluated using the average test-set Uhlmann fidelity between the true and reconstructed density matrices. Results are reported as a function of the number of training states N for a fixed noise strength $\sigma = 0.564$. This choice of presentation allows the data efficiency and scaling behaviour of the learning-based reconstruction methods to be examined in a controlled manner. Figure 1 shows the performance of the fully connected (FC) and convolutional neural network (CNN) models trained using a fidelity-based loss, while Figure 2 shows the corresponding results obtained using a mean-squared error (MSE) loss. In both cases, results are compared against the classical Stokes reconstruction baseline.

Across all dataset sizes considered, the Stokes reconstruction consistently achieves the highest average fidelity. This behaviour is expected in the present setting: Stokes reconstruction directly inverts the known measurement model and therefore does not rely on learning or generalisation across a distribution of quantum states. In the low-dimensional two-qubit regime considered here, the linear inversion problem is well conditioned, and the classical method is able to exploit exact knowledge of the measurement operators. As a result, Stokes reconstruction provides a strong and difficult-to-surpass baseline.

Both neural network models exhibit lower average fidelities than the Stokes baseline, with reconstruction quality improving gradually as the size of the training dataset increases. This trend reflects the data-driven nature of the learning-based approaches: with more training examples, the networks are able to better approximate the inverse mapping from noisy measurement data to quantum states. The fully connected network generally outperforms the CNN at smaller training set sizes, suggesting that the additional inductive bias imposed by the convolutional architecture does not provide an immediate advantage in the low-data regime.

At larger values of N, the CNN exhibits more stable scaling behaviour, particularly when trained using a fidelity-based loss. This may indicate that, once sufficient training data is available, the convolutional architecture is able to extract more consistent features from the structured measurement grid. However, the observed improvements remain modest, and the CNN does not outperform the fully connected model or the classical baseline in the parameter regime explored here.

A clear dependence on the choice of loss function is observed. Training with a fidelity-based loss leads to smoother and more consistent improvements in test fidelity as N increases, whereas the MSE-trained CNN exhibits noticeable fluctuations for intermediate dataset sizes. This behaviour highlights the importance of aligning the training objective with a physically meaningful performance metric. While MSE provides a convenient optimisation target, it does not directly reflect physical state similarity and may encourage the network to prioritise element-wise accuracy over globally relevant corrections.

Despite these differences, neither neural network architecture surpasses the Stokes reconstruction baseline in the experiments reported here. This result suggests that generic

neural network architectures are not yet able to fully exploit the structure of the tomographic measurement process in this setting. In particular, the measurement grid used as input to the CNN reflects a modelling choice rather than an intrinsic notion of locality, and generic convolutional kernels may not align naturally with the underlying physics of the measurement operators.

Overall, these results establish a clear and reproducible baseline for learning-based quantum state reconstruction under measurement noise. They demonstrate that neural networks can learn physically valid reconstructions and improve with increasing training data, while also highlighting the limitations of generic architectures. These observations provide guidance for future work aimed at incorporating more explicit physical structure into neural network designs.

B. Discussion and limitations

The results demonstrate that, while neural network–based reconstruction methods can successfully learn a mapping from noisy tomographic measurement data to physically valid density matrices, their performance remains inferior to that of classical linear inversion for the system sizes and noise levels considered in this study. In particular, the Stokes reconstruction consistently achieves higher average fidelities across all training dataset sizes explored. This indicates that, in the low-dimensional two-qubit regime and under the specific noise model employed here, generic neural network architectures are not yet able to outperform established classical reconstruction techniques.

This outcome is not unexpected. Linear inversion methods such as Stokes reconstruction exploit exact knowledge of the measurement model and do not rely on statistical learning or generalisation across a distribution of states. In contrast, the neural networks must learn an approximate inverse mapping from data that depends on both the underlying state ensemble and the noise model. In the present setting, where the inversion problem is relatively small and well conditioned, this learning task provides limited opportunity for generic architectures to gain an advantage over direct inversion.

The relative performance of the fully connected and convolutional architectures also highlights important limitations. While the CNN exhibits more stable scaling behaviour at larger training set sizes when trained with a fidelity-based loss, it does not surpass the fully connected model or the classical baseline. This suggests that the inductive bias imposed by standard convolutional kernels—namely locality and weight sharing—does not align naturally with the structure of the tomographic measurement grid used here. As discussed earlier, locality in the 6×6 measurement grid reflects a modelling choice rather than an intrinsic physical property, and neighbouring grid entries do not necessarily correspond to closely related measurement operators.

It is also important to recognise limitations imposed by the simulation framework itself. Measurement data is represented at the level of idealised probabilities rather than finite coincidence counts, and noise is introduced exclusively through independent mis-rotations of the measurement operators. While this model captures a physically meaningful class of experimental imperfections, it does not include other relevant effects such as finite sampling noise, detector dark counts, or state preparation errors. The absence of these effects may reduce the relative advantage of learning-based methods, which are often argued to be most useful in complex, noisy, or poorly characterised experimental environments.

Despite these limitations, the neural network approaches considered here possess several advantages that are not fully captured by average fidelity alone. Once trained, the networks provide extremely fast inference, producing reconstructed density matrices via a single forward pass without the need for iterative optimisation. This contrasts sharply with optimisation-based methods such as MLE, which require a separate, potentially expensive optimisation procedure for each reconstruction. As a result, learning-based approaches may be particularly attractive in scenarios involving high-throughput tomography, real-time monitoring, or larger system sizes where classical methods become computationally prohibitive.

Overall, the results presented in this interim report should be viewed as establishing a baseline rather than providing a definitive assessment of the potential of learning-based quantum state tomography. They demonstrate both the feasibility of neural network reconstruction with enforced physical constraints and the limitations of applying generic architectures without incorporating additional physical structure. These observations motivate the future work discussed below, which aims to address the identified limitations by introducing more physically informed modelling choices.

C. Future work

Several directions for future work naturally arise from the results and limitations identified in this interim study. These directions fall broadly into three categories: the incorporation of physical structure into neural network architectures, extensions of the physical noise model, and a more comprehensive assessment of practical performance metrics.

a. Physics-informed network architectures. The most immediate avenue for improvement concerns the design of neural network architectures that more explicitly reflect the structure of the tomographic measurement process. In the present work, the convolutional neural network employs generic kernels originally developed for image-processing tasks, implicitly assuming locality and translation invariance in the 6×6 measurement grid. However, in quantum state tomography, neighbouring grid elements do not necessarily correspond to physically similar measurements unless the ordering is chosen carefully.

A natural extension is therefore to design convolutional kernels or connectivity patterns that encode known relationships between measurement projectors. For example, kernels could be constructed to couple projectors corresponding to the same single-qubit basis on one subsystem, or to reflect symmetries of the underlying measurement scheme. More generally, graph-based neural network architectures may provide a more natural framework for representing tomographic data, where nodes correspond to measurement operators and edges encode physically meaningful relationships between them. Exploring such physics-informed inductive biases represents a central direction for future work.

b. Training strategies and regularisation. Additional improvements may be achieved through systematic exploration of training strategies and regularisation techniques. While the present work uses relatively standard optimisation settings, more careful tuning of learning rates, batch sizes, and regularisation parameters may improve training stability and generalisation, particularly when using fidelity-based loss functions. Techniques such as L_1 and L_2 regularisation, dropout, or early stopping could be investigated to mitigate overfitting and reduce sensitivity to noise.

Architectural depth also warrants further investigation. Deeper networks with fixed parameter budgets may be able to capture more complex nonlinear relationships in the tomographic data without significantly increasing computational cost. Hybrid training strategies, such as pretraining with an MSE loss followed by fine-tuning with a fidelity-based objective, may also help balance optimisation stability with physical relevance.

c. Extensions of the noise and data model. From a physical modelling perspective, extending the noise model to include additional experimental imperfections would increase the realism of the simulations and provide a more stringent test of learning-based reconstruction methods. In particular, incorporating finite coincidence count statistics would introduce shot noise, while detector dark counts and imperfect state preparation would further complicate the reconstruction task. These effects are commonly encountered in real photonic experiments and may provide regimes in which learning-based methods offer clearer advantages over classical techniques.

Related to this, more sophisticated treatments of state preparation and measurement (SPAM) errors could be explored. For example, Palmieri *et al.* demonstrated that neural networks can be used as a preprocessing step to mitigate SPAM errors prior to state reconstruction, leading to improvements in average reconstruction fidelity of order 10% [4]. Integrating such approaches into the present reconstruction pipeline, either as a separate denoising stage or as part of an end-to-end learning framework, represents a promising direction for future investigation.

d. Practical performance considerations. Finally, while the present interim report focuses primarily on reconstruction fidelity, a more comprehensive assessment of practical performance would also consider inference time and computational cost. One of the principal advantages of learning-based reconstruction methods is their ability to produce reconstructions via a single forward pass once trained. Quantifying this advantage through direct comparisons of inference time between neural network–based reconstruction and optimisation-based methods such as MLE would provide a more complete picture of the trade-offs involved, particularly for applications involving high-throughput or real-time tomography.

In summary, although the present results do not yet demonstrate superior reconstruction fidelity compared to classical methods, they establish a robust and extensible framework for benchmarking learning-based quantum state tomography. By identifying the limitations of generic neural network architectures and clarifying the role of physical structure in the reconstruction problem, this work provides a clear foundation for future studies aimed at developing more physically informed and practically effective neural network approaches.

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